



PAPER

C



2018 MATHEMATICS

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

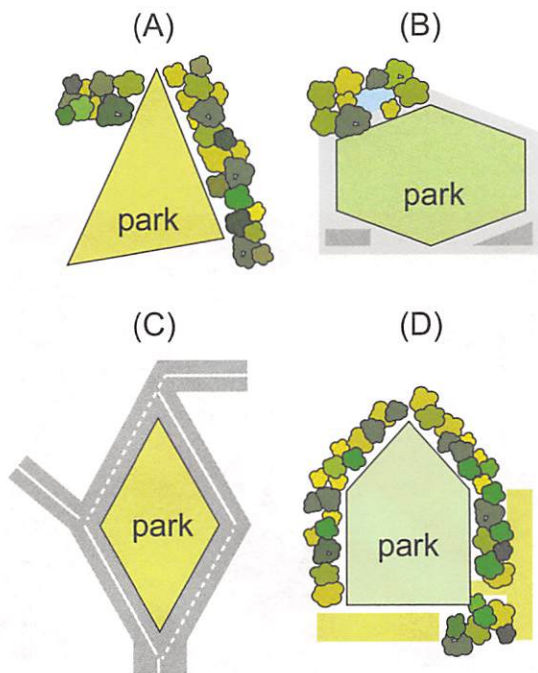
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

ICAS

1. Which park has angles that are all different?



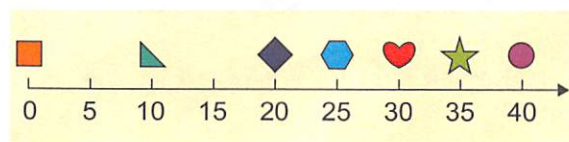
2. Dan's welcome mat is placed on the identical square tiles at his front door.



How many tiles does this mat cover?

- (A) 24 (B) 28
(C) 32 (D) 38

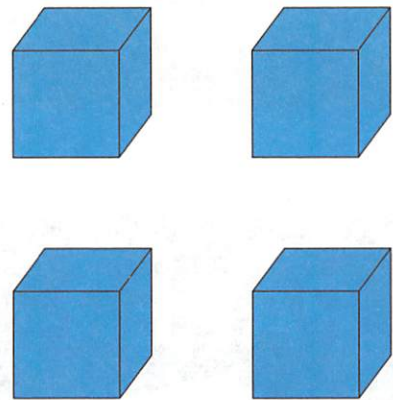
3.



Which shape is halfway between  and  ?

- (A)  (B) 
(C)  (D) 

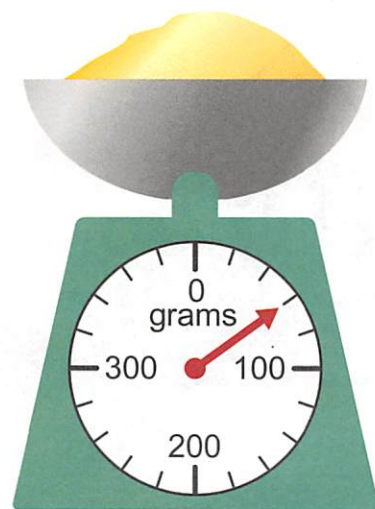
4. Jim wants to join these four identical cubes together to make a new solid.



What type of solid can Jim make?

- (A) a pyramid
(B) a cube
(C) a sphere
(D) a prism

5. John is weighing custard powder using this scale.

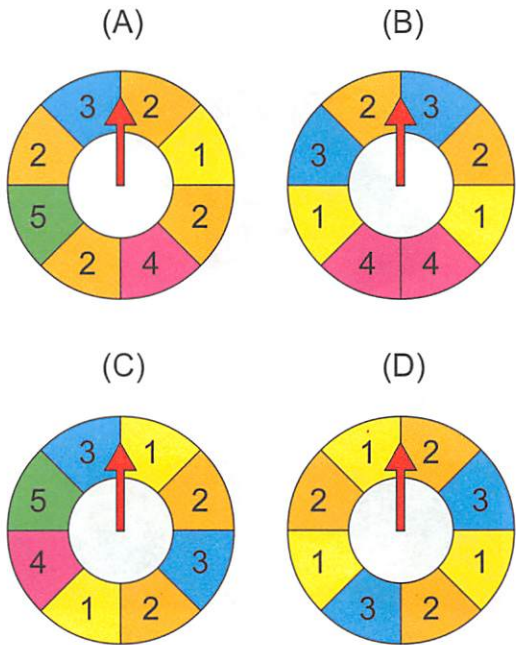


What is the mass of the custard powder?

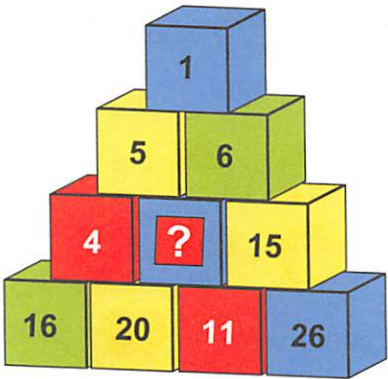
- (A) 3 grams
(B) 6 grams
(C) 30 grams
(D) 60 grams

6. Kate's favourite number is 2.

Which spinner will give Kate the best chance of landing on a 2?



8. This is a picture of Jim's tower.



The number on each block is equal to the difference between the numbers on the two blocks it rests on.

What number must **?** be?

- (A) 1
(B) 9
(C) 11
(D) 19

7. Four children collected shells on the beach. They put their shells in piles as shown.

Child	Number of shells collected	Number of shells in each pile
Tess	27	3
Jade	40	5
Lee	18	6
Sam	32	4

Who made the largest number of piles?

- (A) Tess
(B) Jade
(C) Lee
(D) Sam

9. Lin circled the date her baby brother or sister was due to be born.

It was 9 days until the due date.

September 2018						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6

When did Lin circle the date on her calendar?

- (A) 10 September
(B) 14 September
(C) 23 September
(D) 2 October

10. Helen is solving a puzzle involving a code.

The shapes 😊 and ❤️ represent different numbers.

Clue 1: 😊 + 😊 = 6

Clue 2: 😊 + ❤️ + ❤️ = 7

What is the value of ❤️?

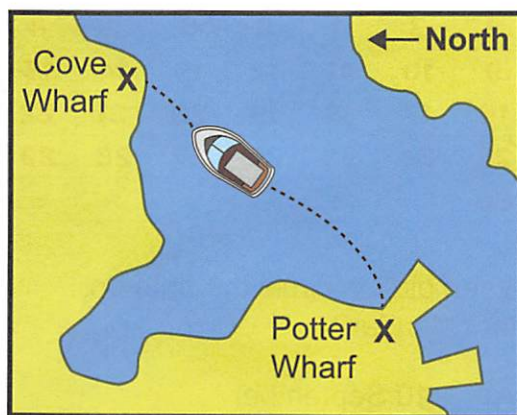
- (A) 4 (B) 3
(C) 2 (D) 1

11. Mrs Liang had 15 oranges. She cut each orange into quarters and gave an equal number of quarters to each of her 5 students.

How many orange quarters did each student get?

- (A) 3
(B) 6
(C) 10
(D) 12

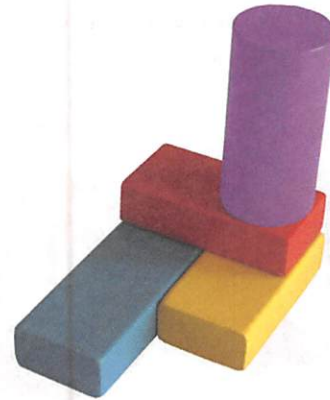
12. A ferry travelled from Potter Wharf to Cove Wharf.



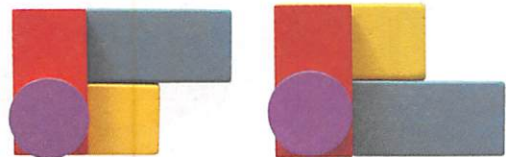
In which direction did the ferry travel?

- (A) north-west
(B) north-east
(C) south-west
(D) south-east

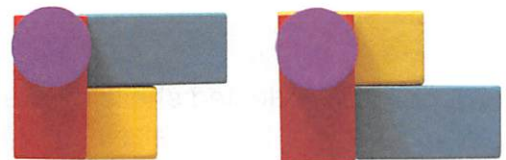
13. Sally made this castle using 4 solids.



Which of these is a view of Sally's castle from above?



- (A) (B)



- (C) (D)

14. Which of the following expressions is **NOT** equal to 276?

- (A) $100 + 50 + 50 + 25 + 25 + 25 + 1$
(B) $100 + 100 + 100 - 10 - 10 - 6$
(C) $100 + 100 + 50 + 20 + 6$
(D) $300 - 20 - 4$

15. Class 4R cleaned up after the carnival.

The table shows the number of items they recycled.

Recycled item	Number
Bottles	
Cans	?
Paper cups	90
Total	200

One-quarter of the items were bottles.

How many cans did the class collect?

- (A) 50 (B) 55
(C) 60 (D) 110

16. Pete wants to post presents to his son for his birthday.

He can choose from these items.



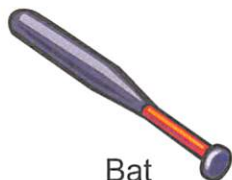
Baseball
150 grams



Helmet
1450 grams



Glove
390 grams



Bat
450 grams

Which items can Pete post so that the total mass is as close as possible to 2000 grams?

- (A) all 4 items
(B) the helmet and the bat
(C) the baseball, bat and glove
(D) the helmet, baseball and glove

17. Arun is at a theme park waiting in a queue for the Wild Jet ride.

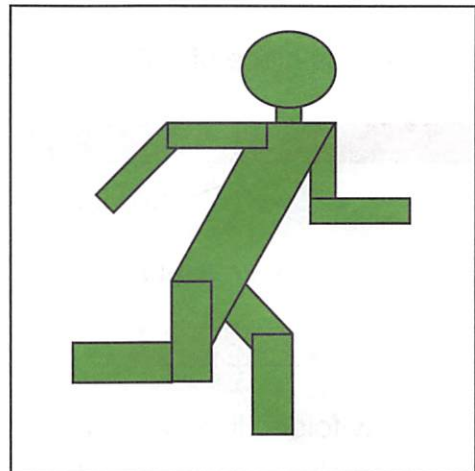
Every 3 minutes, a boat with 5 people leaves.

A boat leaves and Arun counts 26 people ahead of him in the queue.

How many minutes will Arun have to wait until his boat leaves?

- (A) 15 (B) 18
(C) 21 (D) 27

18. Harry made this logo for an exit sign using plane shapes.

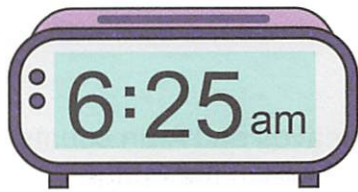


Some shapes overlap.

Which shape was **NOT** used to make the logo?

- (A) 
- (B) 
- (C) 
- (D) 

19. Anna looked at the clock as she left home for her morning swimming training.

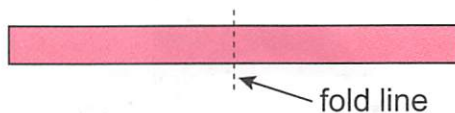


She finished 1 hour 40 minutes later.

What time did Anna finish?

- (A) 8:05 am
- (B) 7:65 am
- (C) 7:05 am
- (D) 6:65 am

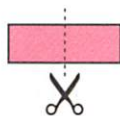
20. Betty had this piece of ribbon.



She folded it in half as shown.



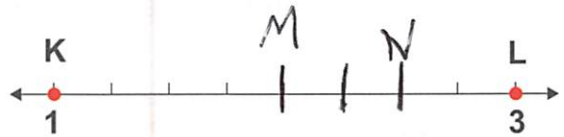
Then Betty folded it in half again and cut the folded ribbon in the middle as shown.



How many pieces of ribbon this length  did Betty have?

- (A) 0
- (B) 2
- (C) 3
- (D) 4

21. Sam plotted two points, K and L, on the number line.



He plotted M halfway between K and L.

Then he plotted N halfway between L and M.

What number is halfway between M and N?

- (A) $1\frac{3}{4}$
- (B) $2\frac{1}{4}$
- (C) $2\frac{1}{2}$
- (D) $2\frac{3}{4}$

22. Priya and Jim started counting from 102.

Priya subtracted the same number each time while Jim's pattern used addition.

Priya: 102, 95, , 81, 74,

Jim: 102, 111, , 129, 138, ...

What is the difference between the values of  and  ?

- (A) 16
- (B) 28
- (C) 32
- (D) 34

23. Min had some cards.

He gave half of the cards to Bill.

Bill gave one third of those cards to Sam.

Sam returned 2 cards to Min and kept 6 cards.

How many cards did Min have in the end?

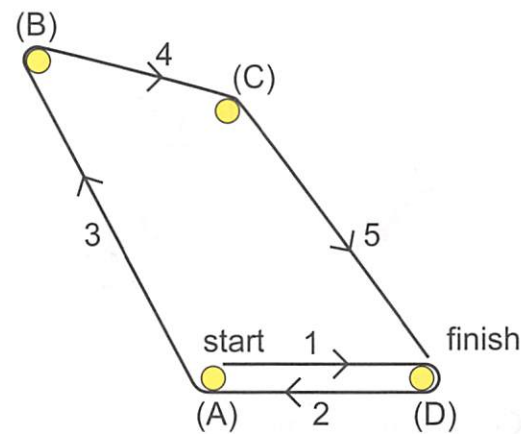
- (A) 20
- (B) 26
- (C) 38
- (D) 50

24. Pamela is in a race.

There are four cones at A, B, C and D.

Pamela starts next to the cone at A, follows the arrows on the diagram and finishes near the cone at D.

Around which cone does Pamela make the greatest change in direction?



25. Ms Clark asked 36 students to choose between ball games and computer games.

She started to put her results in a table.

		Preferred game	
		Computer	Ball
Boys	9		
Girls		?	21
		19	

How many girls preferred ball games?

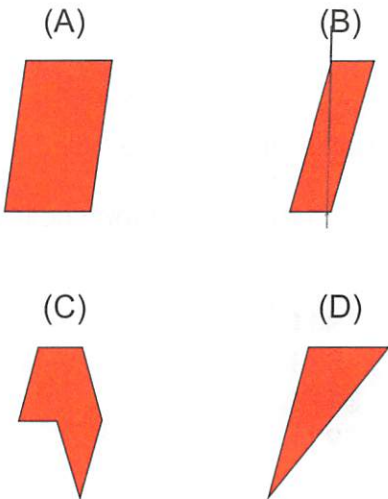
- (A) 8
- (B) 10
- (C) 11
- (D) 13

26. Kevin cut this triangle along its axis of symmetry.



He flipped, slid or turned one of the pieces to fit next to the other piece.

Which of these could be Kevin's new shape?



27. Samir is buying an ice-cream with two different flavours. He can choose to have the ice-cream in a cup or a cone.



How many possible choices does Samir have?

- (A) 6
- (B) 8
- (C) 12
- (D) 16

28. Mr Tin's class is selling cordial at a fair.



They have 7 jugs full of cordial.
Each jug of cordial can fill 4 bottles.
Each bottle of cordial can fill 8 cups.

How much money will they collect if they sell all of the cordial?

- (A) \$44
- (B) \$88
- (C) \$112
- (D) \$224

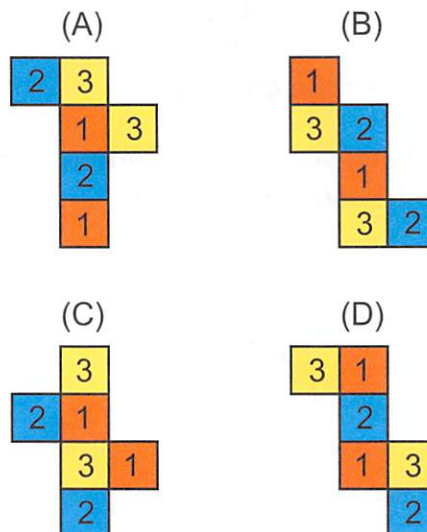
29. The Hui family caught the train from Craven to Anders. This is the train timetable they followed.

Train timetable		
Town	Date	Time
Depart Craven	19 April	8:00 am
Arrive Bendigo	20 April	10:00 am
Depart Bendigo	20 April	11:00 pm
Arrive Anders	21 April	5:00 pm

How long was the train trip, not including the stop at Bendigo?

- (A) 20 hours
- (B) 32 hours
- (C) 44 hours
- (D) 57 hours

30. Which of these nets can be used to make a cube where adjoining faces do **NOT** have the same number on them?



31. Consecutive numbers, such as 6, 7 and 8, follow one after the other.

Five consecutive **EVEN** numbers have a sum of 200.

What is the largest of these five numbers?

- (A) 48
- (B) 44
- (C) 42
- (D) 40

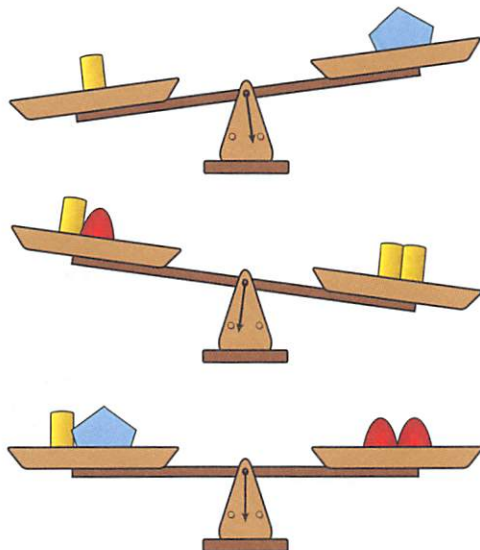
32. Millie bought 15 jars for her homemade jam. The shop charged \$8 for the first 4 jars and a lower fixed price for each jar after that.

Number of jars	5	6	8	15
Cost	\$9.50	\$11	\$14	?

How much did Millie pay for 15 jars?

- (A) \$24.50
- (B) \$25
- (C) \$28.50
- (D) \$29

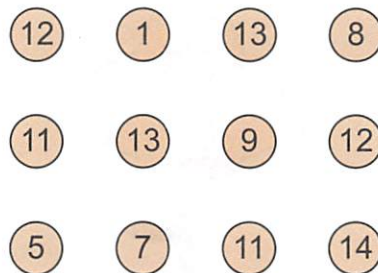
33. Ravi used a balance to compare the masses of some objects. His results are shown below.



Which of the following must be true?

- (A) (B) (C) (D)

34. Jon is solving a puzzle. He can join any circle to the one next to it by using a horizontal or a vertical line.

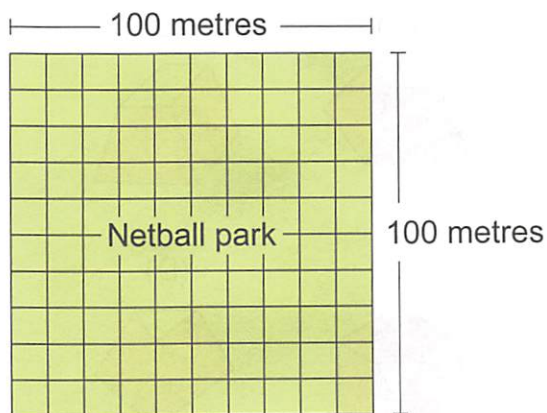


Jon joined 4 circles in this way. He then added the numbers inside the 4 circles together.

What is the largest sum Jon could make?

- (A) 45
(B) 47
(C) 48
(D) 50

35. The local council allocated this piece of land for a netball park.

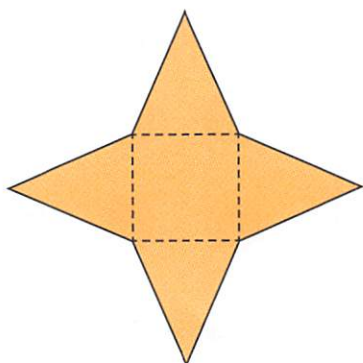


The required area for each netball court, including the space around it, is 35 metres by 20 metres.

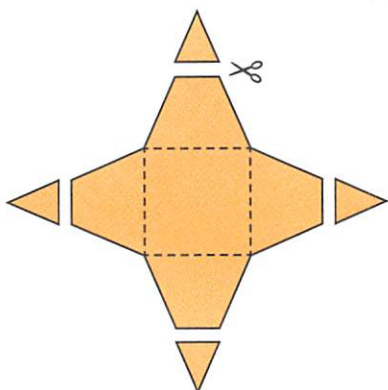
What is the largest number of netball courts that will fit in the park?

- (A) 10 (B) 12
(C) 14 (D) 15

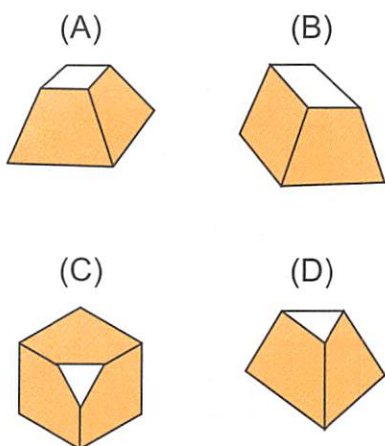
36. Raj is making a 3-D shape from this net.



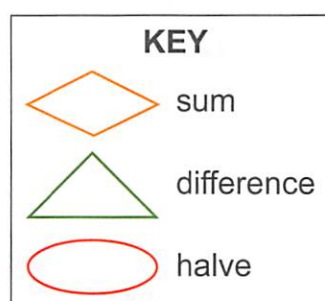
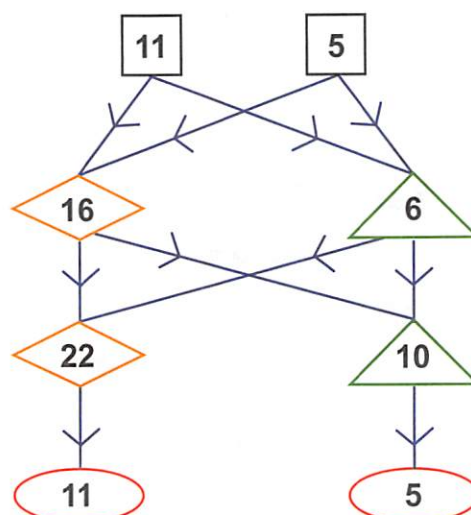
He cut off 4 identical triangles, then folded the net along the dotted lines.



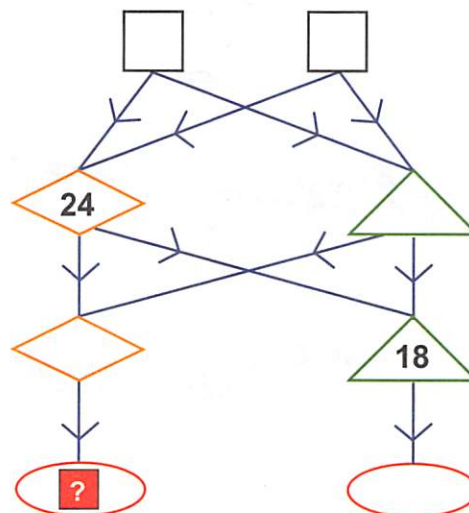
Which of these could be Raj's 3-D shape?



37. Sue followed the instructions in this chart.



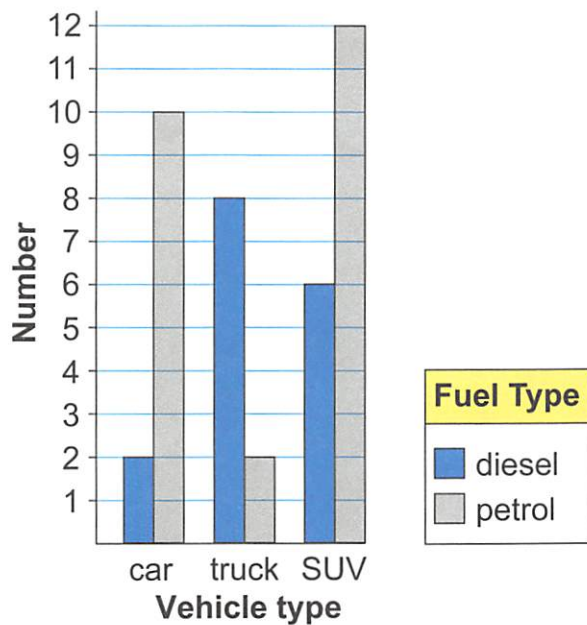
She chose another two numbers and followed the instructions. Some results are missing.



What value must **?** be?

- (A) 9
- (B) 12
- (C) 15
- (D) 21

38. Ben collected data on the vehicles at a used car lot and graphed his results.



Which statement about the vehicles in the car lot is **NOT** true?

- (A) He collected data on 40 vehicles.
- (B) One-fifth of the cars use diesel.
- (C) One-third of the SUVs use diesel.
- (D) Two-fifths of the vehicles use diesel.

39. Paul had a square piece of paper.

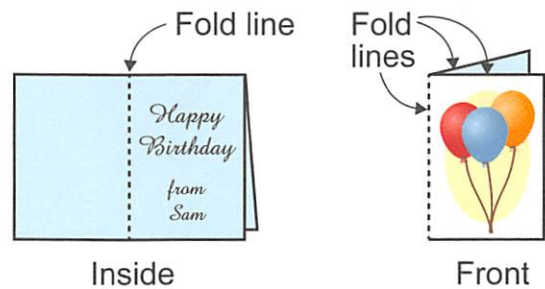
He cut out a rectangle from the paper that was 4 centimetres by 9 centimetres.

The area of the left over paper was 108 square centimetres.

What was the length of the sides of Paul's original square?

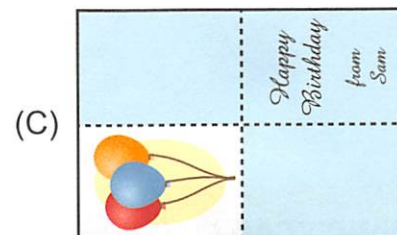
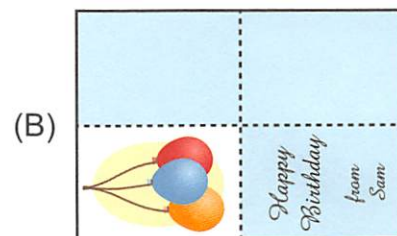
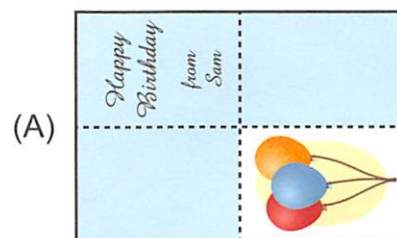
- (A) 9 centimetres
- (B) 11 centimetres
- (C) 12 centimetres
- (D) 13 centimetres

40. Sam folded a rectangular piece of paper in quarters to make this birthday card.



He unfolded the card.

Which one of these is Sam's unfolded card?



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THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

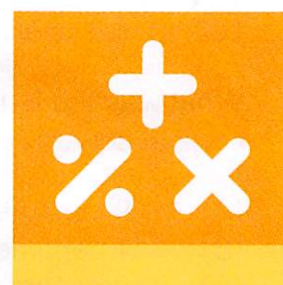
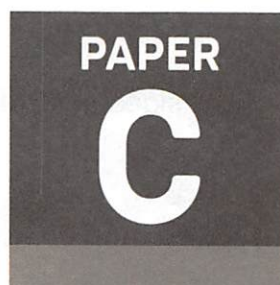
Australia	Year 5
Brunei	Primary 5
Egypt	Year 5
Hong Kong	Primary 5
Indian Subcontinent¹	Class 5
Indonesia	Year 6
Malaysia	Standard 5
Middle East²	Class 5
New Zealand/Pacific³	Year 6
Singapore	Primary 4
Southern Africa⁴	Grade 5

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria, Jordan and Pakistan.

³ Pacific Region: Vanuatu, Papua New Guinea and Fiji.

⁴ Southern Africa Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.



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Answers

Question Number	Area Assessed	Description	Correct Answer
1	Space & Geometry	Identify the shape with angles that are all different	A
2	Measures & Units	Find the area of a shape covering part of a grid	B
3	Number & Arithmetic	Use a number line to identify a half-way point	A
4	Space & Geometry	Determine the type of solid formed by joining identical cubes	D
5	Measures & Units	Read a scale with labels every five markers to measure a mass	D
6	Chance & Data	Identify the spinner with the best chance of landing on a 2	A
7	Number & Arithmetic	Find the largest answer to a number of simple divisions	A
8	Algebra & Patterns	Recognise a pattern with a difference of two numbers	B
9	Measures & Units	Interpret a date on a calendar	B
10	Algebra & Patterns	Solve a problem using substitution into a number sentence	C
11	Number & Arithmetic	Solve a problem involving quarters and sharing	D
12	Space & Geometry	Identify a direction on a map	B
13	Space & Geometry	Identify a solid from a different point of view	D
14	Number & Arithmetic	Identify the expression that is not equivalent to a given value	B
15	Chance & Data	Complete a table that has missing values	C
16	Measures & Units	Determine the objects which have a total mass closest to a given mass	D
17	Number & Arithmetic	Solve a problem involving time and multiplication by three and five	B
18	Space & Geometry	Recognise 2-D shapes in different orientations	A
19	Measures & Units	Solve a problem involving addition of digital times	A
20	Measures & Units	Compare shapes based on length	C
21	Number & Arithmetic	Locate a mixed fraction on a number line	B
22	Algebra & Patterns	Calculate the difference between two terms missing from two sequences	C
23	Number & Arithmetic	Solve a problem involving multiplication and division	B
24	Space & Geometry	Identify the angle in a race which involves the greatest change in direction	D
25	Chance & Data	Calculate the missing numbers in a two-way table	D
26	Space & Geometry	Use the axis of symmetry to create a new shape	B
27	Chance & Data	Use a strategy to find the number of possible outcomes	C
28	Number & Arithmetic	Use multiplication facts to solve a financial problem	C

29	Measures & Units	Solve a problem involving am and pm time	C
30	Space & Geometry	Identify the net that forms a specific cube	D
31	Number & Arithmetic	Solve a problem involving addition and division of even numbers	B
32	Algebra & Patterns	Complete a two-step number pattern	A
33	Measures & Units	Apply logic to solve a balance problem	A
34	Number & Arithmetic	Use a strategy to find the largest possible sum	C
35	Measures & Units	Determine the number of smaller rectangles that will fit into a larger square	B
36	Space & Geometry	Identify the solid formed from a given net	A
37	Algebra & Patterns	Find unknown quantities in a flow chart involving addition, subtraction and division	C
38	Chance & Data	Interpret a column graph to solve a problem	B
39	Measures & Units	Solve a problem involving the area and length of rectangles	C
40	Space & Geometry	Visualise the orientations of pictures on an unfolded card	D